Topic 1.5 • Vectors and Motion in Two Dimensions

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: split vectors into components and analyze projectile motion.

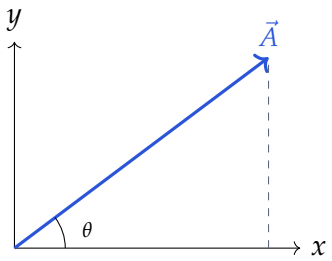
Components

A vector of magnitude A at angle θ above the $+x$ axis:

$A_x =$ _____ $A_y =$ _____

Back: $A =$ _____ $\tan\theta =$ _____

Example: $v = 10\,\text{m/s}$ at 37° ($\cos 37^\circ \approx 0.8$, $\sin 37^\circ \approx 0.6$): $v_x =$ _____, $v_y =$ _____.



Projectile motion

After launch (no air resistance) the two directions are _____:

Horizontal	**Vertical**
$a_x =$ ____: constant v_x	$a_y =$ _____
$\Delta x =$ _____	$\Delta y =$ _____

The _____ is the same in both directions: find it from the vertical motion.

At the top of the path: $v_y =$ ____, but the acceleration is still _____.

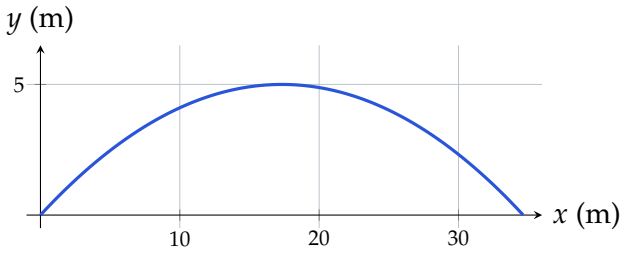
Example 1: horizontal launch

A ball rolls off a table 1.25 m high at 3 m/s.

Time to fall: $1.25 = \frac{1}{2}(10)t^2 \Rightarrow t =$ _____ Lands _____ from the table.

A ball **dropped** from the same height lands _____.

Example 2: launch at an angle



$v_0 = 20\,\text{m/s}$ at 30°: $v_{0x} =$ _____, $v_{0y} =$ _____

Time to the top: _____ Max height: _____

Total time (level ground): _____ Range: _____

Summary